Mingyu Xia

PhD Student at Westlake University

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Google Scholar GitHub

Research Interests

- **Mesoscopic Transport:** Applications of the ***Nonequilibrium Green's Function*** theory to two-dimensional materials.
- **Open Quantum System**: Applying Memory Kernel Coupling Theory (MKCT) to calculate correlation function.

Education

Westlake University, Physics Department — Zhejiang, China

Ph.D. in Science — 08/2025 – 06/2030

- **Supervised by** Wenjie Dou (*Laboratory of Theoretical and Computational Chemistry*)
- **Thesis:** *(Expecting)*
- **Awards:** *(Expecting)* — (Expecting)

Hangzhou Dianzi University, Opto-Electronics Information Science and Engineering — Zhejiang, China

Bachelor of Engineering — 09/2021 – 06/2025

- **Thesis:** *Research & Applications of Micromagnetics Based on LLG Equation* — (Excellent Thesis Exhibition Award)
- **Awards:** The first, second and third prize scholarships once. GPA: 3.83

Publications

[1] M. Xia, W. Dou, *(expecting)*, 2027.

[2] M. Xia, W. Dou, *(expecting)*, 2028.

[3] M. Xia, W. Dou, *(expecting)*, 2028.

[4] M. Xia, W. Dou, *(expecting)*, 2029.

[5] M. Xia, W. Dou, *(expecting)*, 2029.

Experience

APS Global Physics Summit — (Expecting)

Westlake University Teaching Assistant — 03/2026 – 02/2027

- General Physics (Undergraduate Level) — 2026 Spring
- (Doctor Level) — Autumn 2026

Undergraduate Research Summer Camp at CUHK(SZ) — 07/2024 – 08/2024

- School of Science and Engineering, Physics Programme — Academic Exploration Award
- Topic: *Transitions of Magnetic Skyrmion Tube, Toron, and Chiral Bobber Controlled by PMA and External Field in Chiral Magnets*

Honors & Awards

Zhou Peiyuan Mechanics Competition for College Students — 08/2023, 07/2025

- The National Third Prize — Chinese Society of Theoretical and Applied Mechanics

Mathematics Competition of Chinese — 01/2023

- The National Second Prize — Chinese Mathematical Society

Undergraduates Physics Experiment and Department Technological Innovation Competition — 09/2023

- The Provincial Third Prize — Zhejiang Physical Society

Skills

- **IT Skills**: Python, MATLAB, LaTeX, Origin
- **Languages:** Proficient English (IELTS: 6.0) | Semi-skilled Cantonese | Native Chinese